

upon the illuminating quadrant. A, B, C and D are electrical voltages corresponding to laser power falling on the four quadrants.

## Important parameters

Major performance parameters of laser-guided munitions include sensitivity, field-of-view, PRF code compatibility and response linearity. In

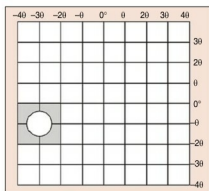


Fig. 5: Operating principle of a two-dimensional array based position sensor

addition, immunity to false codes and response to desired code in the presence of false code are other important parameters relevant to assessing performance efficacy of laser-guided munitions.

Sensitivity is the minimum value of laser power density impinging on the laser seeker cross-section in a plane orthogonal to the optical axis of the seeker head to which it can respond satisfactorily. It is a characteristic of the seeker's front-end optics and photo sensor. Sensitivity of the seeker decides the maximum guidance range for known values of laser target designator parameters, target reflectivity, height of laser target designator (in case of airborne laser designator) and laser seeker above sea surface and visibility conditions.

Field-of-view determines the probability of the weapon finding itself within the laser basket at maximum

guidance range. Laser-guided munitions using a seeker head having larger field-of-view would have a higher probability of finding themselves in the laser basket and subsequently precisely hit the intended target.

PRF code compatibility is the primary requirement for the weapon to function. This means the PRF code of the received laser radiation would be the same as the PRF code programmed in the guidance unit before the start of mission. The code is usually expressed as time interval between two successive laser pulses in milliseconds, usually up to the third decimal place. Two codes are considered compatible if the difference in time periods of the two codes is less than a certain specified value.

Response linearity predominantly decides the circular error probability (CEP). Immunity to false PRF codes and capability to stay locked to the desired code in the presence of false PRF codes enhances the probability of target hit. The former test is performed by irradiating the seeker head with a PRF code different from the programmed PRF code and later by irradiating the seeker head simultaneously with radiations of correct and false PRF codes.

## Deployment configurations

Different deployment configurations are used for laser-guided munitions delivery, depending upon the nature of mission. The one illustrated in Fig. 2 uses a ground based laser target designator and aerially-delivered bomb. There can be other possible scenarios. For example, in another case, the laser target designator is located on another aircraft, which is not the one carrying the laser-guided bomb. Such a deployment scenario is depicted in Fig. 6.

In yet another deployment configuration, the target is designated from a ground based designator and guided munitions are also launched from a ground based platform. This is true in case of cannon-launched laser-guided projectiles. This is depicted in Fig. 7.

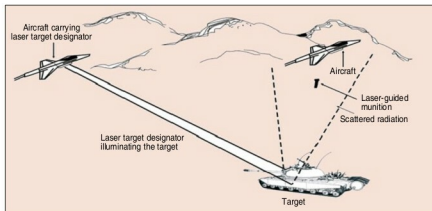


Fig. 6: Laser-guided munitions delivery with target designated from another aircraft

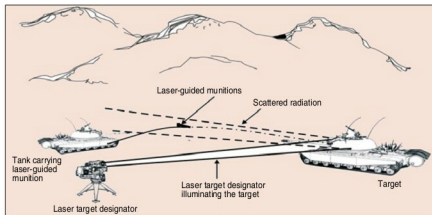


Fig. 7: Laser target designator operation and laser-guided munitions delivery from ground based platforms